Palaeomagnetic Investigations of the Tertiary and Quaternary Igneous Rocks: VII The Tertiary Rocks of Southwest Germany

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With 4 figures and 2 tables
and a note added in proof

Zusammenfassung

Aus dem Bereich des Oberrheingrabens und des Hegaus (SW-Deutschland) wurden Vulkanite entnommen und paläomagnetisch und erzmineralogisch untersucht.

Die umfangreiche Gesteinsserie aus dem Hegau zeigt, daß zwei datierte Gesteine der von Heirtzler et al. (1968) vorhergesagten Magnetisierungsrichtung ent- und zwei dieser Skala widersprechen. Dabei bleibt offen, inwieweit die beiden Skalen radiometrischer Daten übereinstimmen. Möglicherweise gibt die gefundene Remanenz in den Katzenbuckelgesteinen die Zeit der hydrothermalen Veränderung, der Rest das paläomagnetische Feld wieder.

Abstract

Rocks from sixteen sites in southwestern Germany were examined palaeomagnetically and petrologically with attention concentrated on the ore minerals. The largest suite of rocks from the Hegau shows that two dated rocks have the sign of magnetization predicted by Heirtzler et al. (1968) and two do not. However the confidence levels of the radiometric ages are not known. It is probable that the remanence found in the Katzenbuckel reflects the time of hydrothermal alteration, the rest, the paleomagnetic field.

Résumé

Des échantillons volcaniques, prélevés dans la région du fossé d'Alsace et de Bade ainsi que dans celle du Hegau (Allemagne du sud-ouest) ont fait l'objet d'une analyse paléomagnétique et minéralogique.

La série de roches provenant du Hegau montre que 2 roches datées correspondent à la direction d'aimantation indiquée par Heirtzler et d'autres (en 1968), tandis que 2 autres la contredisent. La question de savoir dans quelle mesure les deux échelles de données radiométriques sont concordantes reste entière. Il se peut que l'aimantation rémanente, dans les roches du Katzenbuckel, indique le moment de la transformation hydrothermale, tandis que le reste reflète le champ paléomagnétique.

Краткое содержание

Для палеомагнитных и рудноминералогических исследований собрали вулк-а ниты из области верхнерейнского грабена и Hegau (юго-восточная Германия).

По обширной серии пород из Hegau установили, что две породы, датированные Heirtzler'ом и др., (1968), соответствуют предсказанному направлению намагничивания, а две другие стоят в противоречии. При этом вопрос, в какой степени обе шкалы радиометрических данных совпадают, остается открытым. Возможно, что установленный остаточный магнетизм отображает в породах из Katzenbuckel период гидротермальных изменений, а остальные — направление палеомагнитного поля.

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Intruduction

There are relatively few exposures of Cenozoic igneous rocks in southwestern Germany. Those which do occur appear to be related to the Rhine graben structure (e. g. Stellrecht, 1964; Wimmenauer, 1967) with the exception of the Hegau volcanics, although these too are related to faulting patterns. Petrological investigations (Frenzel, 1955; Weinelt et al., 1965 etc.) and age determinations (Lippolt et al., 1963) have been carried out in order to better establish the relationship between tectonic and volcanic history of the Rhine graben. The present study was carried out in order to provide one further parameter, the direction of remanent magnetization. The palaeomagnetic study[1]) was accompanied by investigation of the ore mineralogy with particular reference to the oxidation state of the iron oxides (Ade-Hall et al., 1968; Wilson and Haggerty, 1966). Prior to the present study only the rocks of the Kaiserstuhl had been examined palaeomagnetically (Lauer, 1964, 1967). The volcanic rocks of the Rhine graben are located primarily along the shoulders of the graben the principal exception being the Kaiserstuhl volcano. Geographically six local areas can be distinguished (Fig. 1).

I. The southern margin of the Rheinisches Schiefergebirge (Rheingau), including those volcanic rocks underlying parts of the Mainz Basin (Negendank, 1968, 1969; Ehrenberg, 1968).
II. The Vorspessart region (Weinelt et al., 1965).
III. The Sprendlinger Horst and northern Odenwald (Klemm, 1905).
IV. The Vorderpfalz (Haardtrand) (Stellrecht, 1964).
V. The Kaiserstuhl region.
VI. The region of the Hegau volcanics, some 80—90 km east of the Kaiserstuhl. The area lies beyond the Rhine graben region in a strict sense and is closer to Lake Constance (V. Engelhardt and Weiskirchner, 1961; Leopold, 1940).

Petrological investigation

The principal rock types collected were melilite ankaratrite, nephelinite, tephrite, basanite, phonolite and trachyte. Their petrology has been reviewed by Wimmenauer (1967) as part of the Rhine graben report. Their distribution is shown on the accompanying figure (Fig. 1) which is modified from that publication. The volcanics are of various ages reaching from upper Cretaceous to Pliocene.

The thin section investigation did not provide any new information. As might be anticipated the acid volcanics contained very little or no ore minerals while in the more basic eruptives the content varied from 1—9 Vol.%. The ore minerals with the exception of some rare pyrite flakes, consisted exclusively of titanomagnetite of class 1 (Buddington and Lindsley, 1964; Wilson and Haggerty, 1966; Ade-Hall et al., 1968; Nairn et al., 1971). The titanomagnetite commonly shows a zonal structure with a colour grading from blue gray in the centre to flesh tones at the margins. It is very rare to find grains belonging to oxidation class 2.

[1]) This study was made possible through the support of NATO grant 355 which is gratefully acknowledged.

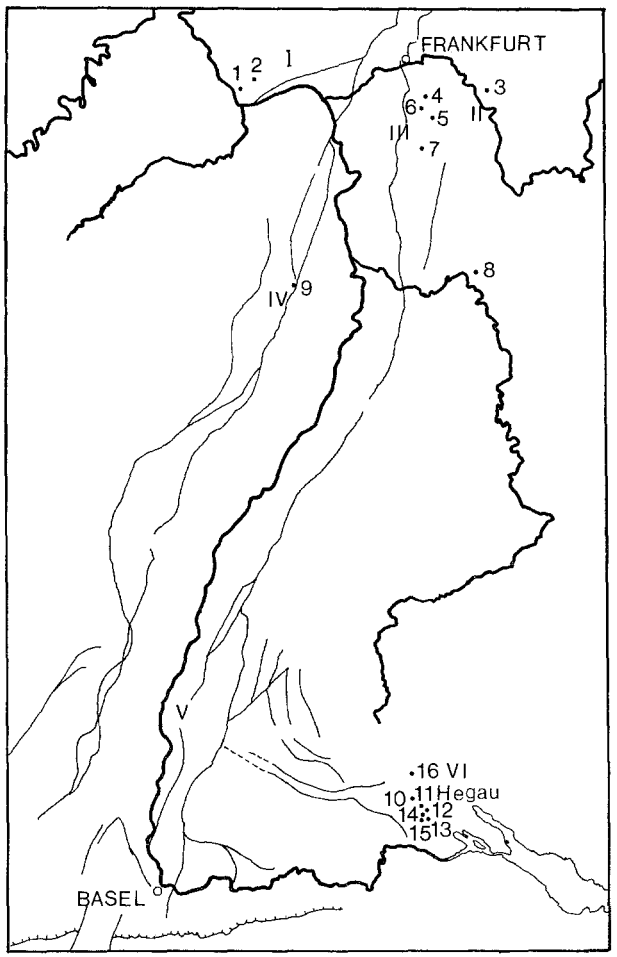


Fig. 1. The location of the Tertiary igneous rocks in Southwest Germany.

Maghemite occurs rarely, in the nephelinite from Ringwäldchen granulation with strong internal reflection was found. Only in the sanidine nephelinite from Katzenbuckel was some haematite found. In general ilmenite and the higher oxidation stages are lacking in marked contrast to rocks from Hungary (NAIRN et al., 1971) or Mexico (NEGENDANK, 1972). The low oxidation state, possibly resulting from the low O_2-fugacity of the magma may be related to the depth of emplacement of the stocks. According to RÜGER (1928) in the region of the Katzenbuckel at least 600 m. of sedimentary cover must have been removed since intrusion.

As in the earlier study of Hungarian rocks (NAIRN et al., 1971) no correlation was found between oxidation state and magnetic sign. Notes describing the ore mineralogy are given in tabular form (table 1) so arranged that they may be compared with the palaeomagnetic data (table 2).

Table 1. **Summary of Petrological Data.**

	Petrology	Reference	Ore Mineralogy
I. Rheingau			
1. Waldburghöhe	basanitic nephelenite	Ehrenberg, 1968	Ores 2.5% by volume, Titanomagnetite class 1 and 2. Rare maghemite along grain margins and cracks. Ores of two generations, titanomagnetites commonly zoned.
2. Rabenkopf	basanitic nephelenite	Ehrenberg, 1968	Ore 9% by volume. Titanomagnetite class 1 rare grains class 2 (small to medium sized grains). Grains vary in size from point size to that of groundmass pyroxenes. Titanomagnetite in olivine is blue-gray coloured.
II. Vorspessart			
3. Rückersbacher Schlucht	trachytoid phonolite	Weinelt et al., 1965	Ores ~ 1%. Titanomagnetite class 1. Ores mostly as phenocrysts. Indeterminate dust like grains also occur.
III. Sprendlinger Horst and Northern Odenwald			
4. Hohenberg (Heusenstamm)	trachyte	Klemm, 1905	Very little ore, too finely divided for identification.
5. Thomashütte (Dietzenbach)	trachyte	Klemm, 1905	No identifiable ores in section examined.
6. Ringwäldchen (Götzenhain)	nephelinite and limburgite	Negendank, 1969 Klemm, 1910	Ores 2—4% uniformly distributed. Titanomagnetite class 1 often with marginal granulation. Grains smaller in size than groundmass, and show zonal structure. Pyrite flakes occur.
7. Rossberg (Rossdorf)	olivine nephelinite and nepheline basanite	Becker, 1904 Klemm, 1910, 1938 Beisler, 1955 Negendank, 1969	Ores 6.5%. Titanomagnetite class 1. Grains size of groundmass pyroxene.

8. Katzenbuckel	sanidine nephelinite	FRENZEL, 1955 LIPPOLT et al., 1963	Ores 15%. Original ore content difficult to determine as mafic minerals are more or less altered to ore minerals. Titanomagnetite class 1 is rare, mostly altered to haematite. Some pyrite. Rock has been hydrothermally altered (FRENZEL, 1955).
IV. Vorderpfalz			
9. Forst	olivine nephelinite	STELLRECHT, 1964 STELLRECHT & EMMERMAN, 1970	Ores 5% in two generations. Titanomagnetite class 1 with a regular zonal structure, the colour grades from blue-gray at centre to flesh colour at the margin.
V. Hegau			
10. Hohenhöwen	nepheline melitite ankaratrite	LEOPOLD, 1940 v. ENGELHARDT and WEISKIRCHNER, 1961	Ores 3%. Large grains, sometimes idiomorphic. Titanomagnetite class 1.
11. Mägdeberg	nosean nepheline phonolite	LIPPOLT et al., 1963	Very little ore, some pyrite flakes. Ore too finely divided for identification
12. Hohenkrähen	nosean phonolite	LIPPOLT et al., 1963	No identifiable ore in section examined
13. Hohentwiel	nosean phonolite	LIPPOLT et al., 1963	One grain of titanomagnetite class 1 recogniced with some maghemite in section examined
14. Gönnersbohl	trachytoid phonolite	LIPPOLT et al., 1963	No identifiable ore in section examined.
15. Staufen	leucite phonolite	CUSHING and WEINSCHENK, 1893	
16. Höwenegg	melitite ankaratrite	LEOPOLD, 1940 v. ENGELHARDT et al., 1961	Ores 6.8%, in xenomorphic ores of various sizes. Titanomagnetite class 1 with zonal structure.

Table 2. Summary of The Palaeomagnetic Data from The Cenozoic Volcanics of S.W. Germany.
(This table was completed by radiometric data from Horn, P., Lippolt, H. J., Todt, W., 1972.)

		Age in Million years	Declina-tion Inclina-tion	Sampling	α_{95}	K	R	Demag. oe.
I. Rheingau								
1. Waldburghöhe	nephelinite stock	40.6	204.2 — 35.6	4/6	8.6	115	3.974	300 oe.
2. Rabenkopf	nephelinite stock		346.9 + 57.8	6/6	9.8	48	5.896	77 oe.
II. Vorspessart								
3. Rückersbacher Schlucht	phonolite	59.8	231.5 — 49.6	5/6	18.0	19	4.789	150 oe.
III. Sprendlinger Horst and Northern Odenwald								
4. Hohenberg, S. of Heusenstamm	trachyte	64.6	262.2 — 43.1	4/7	29.3	11	3.722	1200 oe.
5. Thomashütte near Dietzenbach	trachyte		141.5 — 70.8	7/7	4.7	168	6.964	150 oe.
6. Ringwäldchen near Götzenhain	limburgite (nephelinite)	39.48	356.9 + 69.0	6/6	5.9	131	5.962	300 oe.
7. Rossberg near Rossdorf	olivine nephelinite	42.5	204.1 — 61.2	6/6	8.2	68	5.930	NRM
8. Katzenbuckel	sanidine nephelinite	53—55	23.4 + 49.7	5/8	17.5	20	4.800	77 oe.
IV. Vorderpfalz								
9. Forst	olivine nephelinite	53.3	134.6 — 64.3	7/7	7.4	67	6.910	77 oe.
V. Hegau								
10. Hohenhöwen	nepheline melilite ankaratrite		351.5 + 72.0	7/7	10.1	37	6.837	77 oe.
11. Mägdeberg	nosean nepheline phonolite	8.6	no result					
12. Hohenkrähen	nosean phonolite	8.5	15.4 + 67.7	6/6	5.0	178	5.972	77 oe.
13. Hohentwiel	nosean phonolite	9.5	188.0 — 42.2	6/6	6.3	115	5.960	150 oe.
14. Gönnersbohl	phonolite	9.2	172.3 — 43.4	6/6	34.6	4	5.500	150 oe.
15. Staufen	leucite phonolite		4.0 + 63.9	7/7	9.1	45	6.870	150 oe.
16. Höwenegg	melilite ankaratrite	6.0	356.3 + 76.8	7/7	27.3	5	6.620	NRM

Palaeomagnetic investigation

The sampling and measurement techniques are similar to those described in earlier papers (NAIRN, 1967; NAIRN and VOLLSTÄDT, 1967) so that they will only be briefly outlined here. For logistic reasons it was found more convenient to collect oriented block samples and core in the laboratory than to use a portable field drill. In general approximately six blocks were collected from each site and from these a single cylinder was prepared. At outcrops samples were taken as widely separated as possible to avoid lightning effects. Whenever possible sampling in quarries was preferred to natural outcrop.

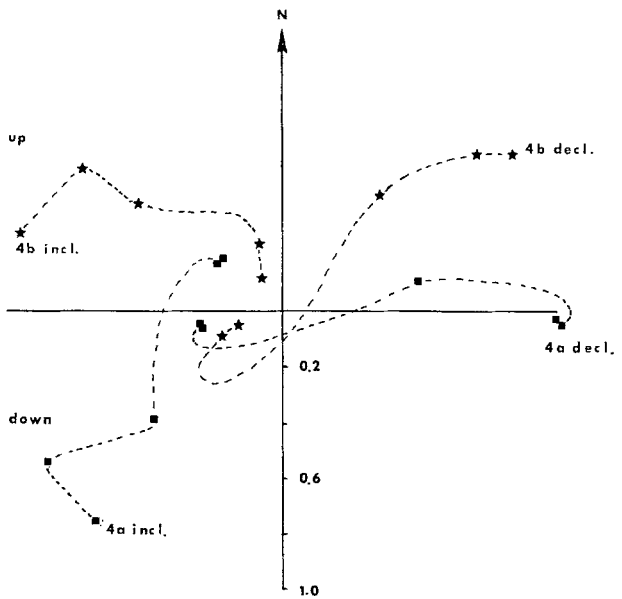


Abb. 2 a

Fig. 2. Plot of magnetic declination and inclination against normalized intensity during alternating field demagnetization. The demagnetizing fields are indicated. The numbers refer to sites and corresspond to those listed in table 2.

All samples were progressively demagnetized in alternating fields. Two samples per site were demagnetized in five stages up to 1200 oe. From a plot of the directions of magnetization against intensity (Fig. 2) it was possible to determine when secondary components of magnetization had been removed. The remainder of the samples from that site were then demagnetized and measured at that stage and one or two higher stages. The natural remanent magnetization was first measured in every case. The measurements were carried out using a Foster fluxgate spinner magnetometer. Some normalized demagnetization curves are illustrated (Fig. 3).

Site mean directions of magnetization were calculated using a Wang computer and the program devised by S. Grommé. The criterion adapted for the best representation of the true or original magnetization acquired when the rock

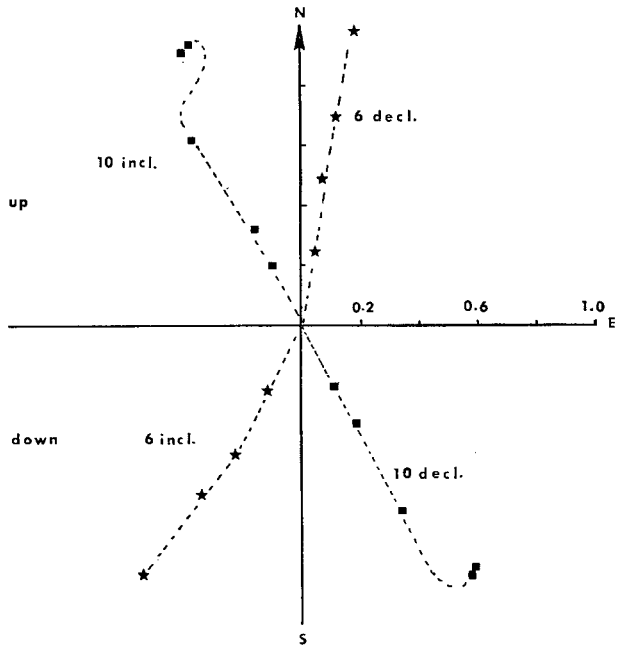


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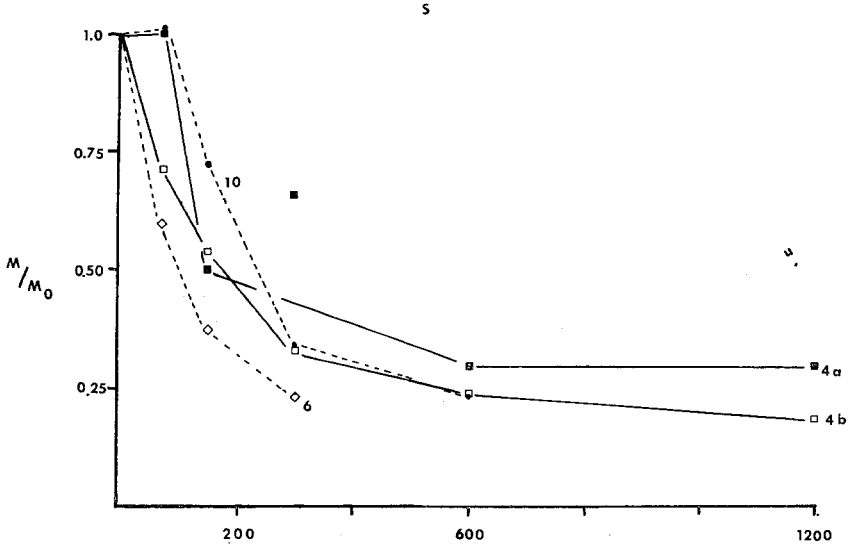


Fig. 3. Normalized demagnetization curves illustrating the change in intensity with increasing fields. The numbers correspond to sites as listed in table 2.

cooled was selected on the basis of minimum circle of confidence and scatter (maximum K). These results are summarized in table 2 and illustrated in Fig. 4.

No representation of the distribution of the intensity of magnetization (either natural remanent magnetization or after demagnetization) is given because of the heterogeneous nature of the collection.

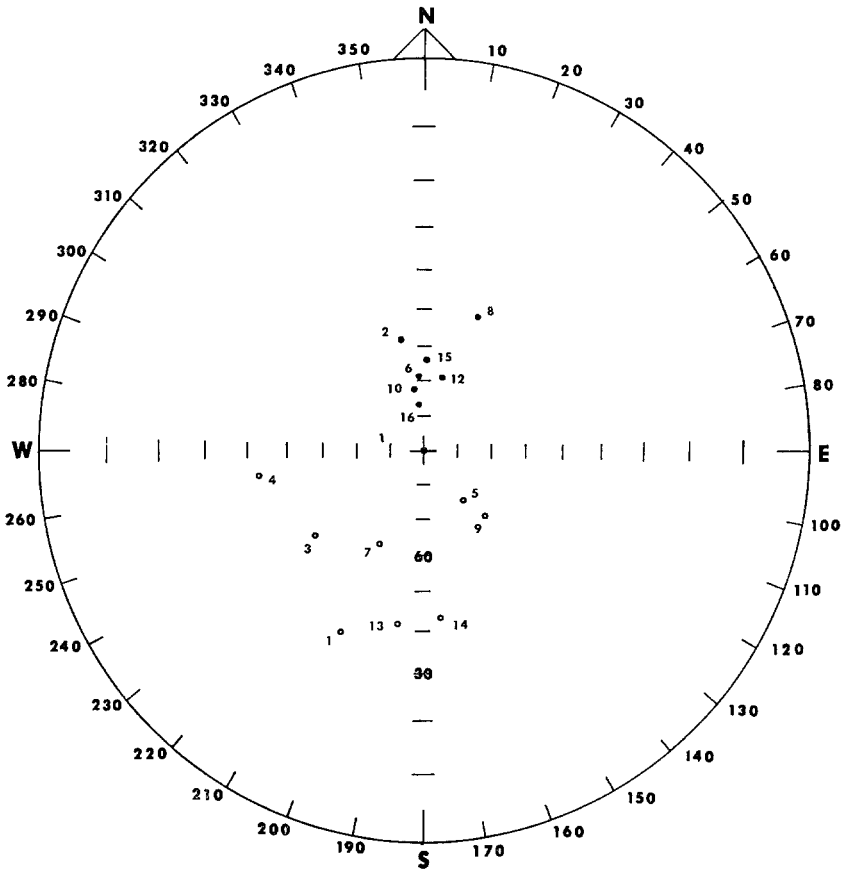


Fig 4. Site mean directions of magnetization of rocks from Southwestern Germany. Site numbers corresspond to those given in table 2. Following standard convention solid symbols represent normal magnetization (north seeking pole pointing downwards).

Discussion of results

There is considerable variation in the quantity of identifiable ore minerals found in the rocks considered. The range is from none identifiable to 9%. In the case of Katzenbuckel the percentage of ores is even higher (15%) but the rock has been hydrothermally altered (FRENZEL, 1955). This means that the palaeomagnetic result is more likely to reflect the time of alteration rather than of emplacement of the stock. There is no obvious correlation between the amount of ore present and the confidence limits of the mean direction of magnetization, however it is apparent that in those rocks where no ore minerals were observed in polished section, the radius of the circle of confidence was small. This may indicate that the remanence is being carried in the finely divided, unidentifiable ore fraction and one might anticipate such rocks exhibit a high coercive force.

A number of the Hegau volcanic rocks have been radiometrically dated

(LIPPOLT et al., 1963) and it is of interest to compare the sign of magnetization of these rocks with that predicted from magnetic anomaly patterns (HEIRTZER et al., 1968). The results are listed below:

Site	Age	Magnetization	HEIRTZLER normal periods
Höwenegg	6.0 my	356.3 + 76.8	5.96—6.24 my
Hohenkrähen . . .	8.5 my	15.4 + 67.7	8.37—8.51 my
Mägdeberg . . .	8.6 my	no result	
Gönnersbohl . . .	9.2 my	172.3 — 43.4	8.79—9.94 my
Hohentwiel . . .	9.5 my	188.0 — 42.2	8.79—9.94 my

While in view of the uncertainties associated with the age determinations there can be no definite conclusion, it is worth noting the conflict between the reversed magnetization of the two oldest rocks and the predicted normal magnetization (anomaly 5) of HEIRTZLER et al. (1968) in view of the duration of that normal period 8.79—9.94 my.

Note added in proof:

Since the present paper was submitted, two relevant articles have appeared: MÄUSSNEST, G., Magnetische Untersuchungen an einigen Förderschloten und Deckentuffen der Hegau-Vulkanprovinz, Oberrhein. geol. Abh., **20**, 103—113, 1971, and HORN, P., LIPPOLT, H.-J., TODT, W., Kalium-Argon-Altersbestimmungen an tertiären Vulkaniten des Oberrheingrabens, Eclog. geol. Helv., **65**, 131—156, 1972.

The first paper reports some magnetic measurements upon rocks from the Hegau. Although not duplicating our sites, there are quite large differences between the remanence measurements on tuffs reported and the measurements we report on volcanic necks (eg. Höwenegg and Höhenhöwen) which are not easily understood since all results reported have been subjected to magnetic sign (HEIRTZLER et al., 1968) as follows:

	Age	Magnetization	HEIRTZLER et al. (1968) predicted magnetic sign
Ringwäldchen (Götzenhain)	39.5 my	356.9 + 69.0	R
Roßberg	42.5 my	204.1—61.2	N
Forst	53.3 my	134.6—64.3	N
Katzenbuckel	53—55 my	23.4 + 49.7	—

While because the uncertainties of the age determinations are long compared with the duration of a polarity interval the comparison cannot be consignificant, it cannot escape notice that the agreement of predicted and observed signs is poor.

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Ferromanganese nodules from the Devonian of the Montagne Noire (S. France) and West Germany

By Maurice E. Tucker, Freetown *)

With 12 figures and 1 table

Zusammenfassung

Es werden Eisenmanganknollen (als Äquivalent zu den rezenten Manganknollen) aus dem oberdevonischen Griotte (rote pelagische Kalke) der Montagne Noir und den Cephalopoden-Kalken des Rheinischen Schiefergebirges beschrieben. Sie treten als Umkrustungen von Skeletten und Kalksteinbruchstückchen auf und zeigen gewöhnlich Kolloform-Strukturen. Die Knollen sind vergesellschaftet mit umkrusteten Foraminiferen, und es wird deshalb sublitorales Bildungsmilieu angenommen. Verglichen mit rezenten sind die oberdevonischen Knollen durch Auslaugung von Mangan relativ Eisen-angereichert, was auf diagenetische Mobilisation zurückzuführen ist. Untersuchungen mit der Mikrosonde haben gezeigt, daß Mangan und Calcium gleichläufig, Mangan und Eisen und Kieselsäure gegenläufig sind. Diagenetische Hämatit-Anreicherungen gibt es im Griotte an hardground-Horizonten; es wurden zwei Mineralisationsphasen festgestellt. Die oberdevonischen Eisenmangan-Knollen zeigen, im Gegensatz zur allgemein verbreiteten Meinung, daß sich die Knollen bei Überlagerung nicht unbedingt auflösen.

Abstract

Ferromanganese nodules (equivalent to Recent manganese nodules) are described from the Upper Devonian griotte (red pelagic limestone) of the Montagne Noire (S. France) and the Cephalopodenkalk of the Rheinisches Schiefergebirge, West Germany. They occur as encrustations, commonly exhibiting colloform structures, around skeletal material and limestone clasts. The nodules are associated with encrusting foraminifera and a development in the sublittoral environment is envisaged. Chemically, the ferromanganese nodules are depleted in manganese relative to iron, compared with Recent nodules, a loss which is attributed to diagenetic migration of manganese. Electron probe studies show that manganese covaries positively with calcium, but

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